

Assignment 2 - Project Documentation

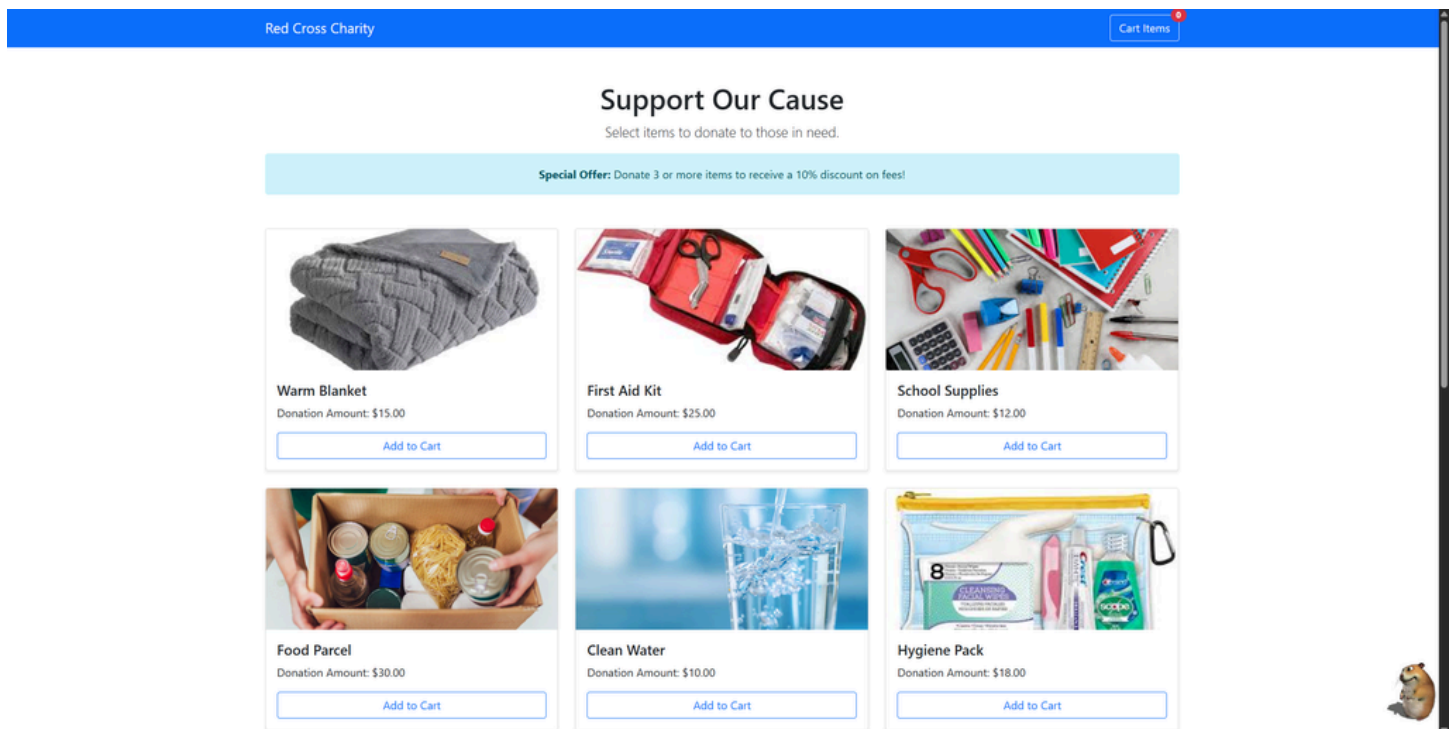
Charitable Online Donation System (Red Cross Charity)

The goal of this project was to create a single-page frontend application for a charitable organization.

The site allows users to browse donation items (such as blankets, food, and medicine), add them to a shopping cart, and proceed to a checkout form.

The project relies on Bootstrap 5 for the responsive design and layout, and JavaScript for the functionality (SPA navigation, cart logic, and form validation).

Theme: A humanitarian aid shop where users "purchase" donations for people in need.



Bootstrap Components Used

1) Navigation Bar

Classes used: .navbar, .navbar-expand-lg, .navbar-dark, .bg-primary, .sticky-top

Location: The header of the page.

It provides branding and holds the "Cart" counter. It is sticky, so it stays visible while scrolling. The brand link also acts as a "Home" button to return to the gallery view.



2) Grid System & Containers

Classes used: .container, .row, .col-md-4, .col-sm-6

Location: The main product gallery (#product-container) and the checkout layout.

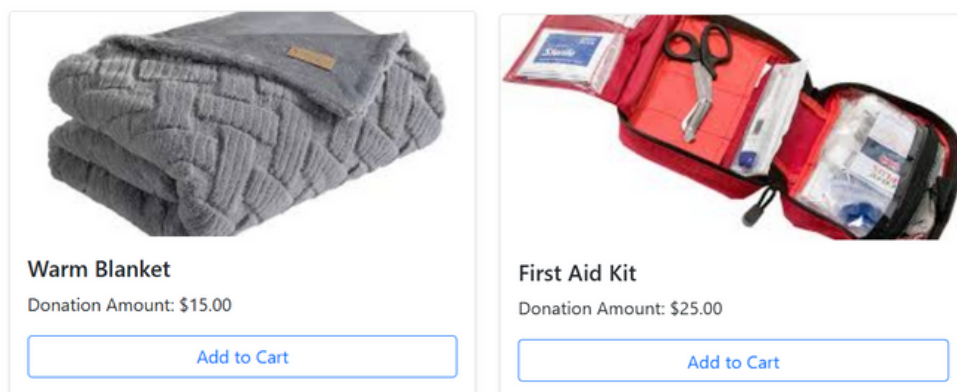
The grid system ensures the product gallery is responsive. It displays 3 items per row on desktop screens and adjusts to 2 or 1 item on smaller mobile screens.

3) Cards

Classes used: .card, .card-img-top, .card-body, .card-title, .shadow-sm

Location: Inside the gallery to display each donation item.

Cards provide a clean container for the product image, title, price, and "Add to Cart" button.



4) Buttons

Classes used: `.btn`, `.btn-primary`, `.btn-success`, `.btn-outline-light`

Location: Throughout the site ("Add to Cart", "Proceed to Checkout", "Confirm Donation").
To provide clear calls-to-action. I used different colors (Primary and Success) to indicate the primary action on the page.

5) Forms and Inputs

Classes used: `.form-control`, `.form-label`, `.form-text`

Location: The checkout section (`#checkout-view`).

These classes style the input fields (Name, Email, Address) to be full-width and modern, with proper spacing and accessible labels.

6) Badges

Classes used: `.badge`, `.rounded-pill`, `.bg-danger`

Location: Inside the Navbar cart button.

Dynamically displays the number of items currently in the user's cart

7) Alerts

Classes used: `.alert`, `.alert-info`

Location: Top of the gallery view.

Notifies the user about the special discount (10% off for 3+ items)

JavaScript Logic

The site functions as a Single Page Application (SPA). Instead of loading new HTML files, JavaScript is used to hide and show specific <div> containers.

A. Navigation (Hiding/Showing Views)

I created three main views in the HTML:

#gallery-view (Product List)

#checkout-view (Form)

#confirmation-view (Receipt)

I used the Bootstrap utility class .d-none to manage this.

Function showCheckout(): Adds .d-none to the gallery and removes it from the checkout section.

Function showGallery(): Reverses this process to go back.

B. Shopping Cart Logic

Data: A global array cart = [] stores the selected item objects.

Adding Items: The addToCart(id) function finds the matching object from the products array and pushes it into the cart array.

UI Update: Every time an item is added, updateCartUI() runs to update the red badge counter in the navbar.

C. Validation

Before processing the donation, the validateAndSubmit() function checks the form inputs:

Empty Fields: Checks if any required field is empty.

Email Format: Uses .includes('@') to ensure the email is valid.

Phone Number: Uses isNaN() to ensure the phone input contains only numbers.

ZIP Code: Checks if the length is greater than 6 characters.

```
// Check inputs before processing
function validateAndSubmit() {
  const name = document.getElementById("fullName").value;
  const email = document.getElementById("email").value;
  const phone = document.getElementById("phone").value;
  const zip = document.getElementById("zip").value;
  const address = document.getElementById("address").value;
  const errorMsg = document.getElementById("error-msg");

  errorMsg.innerText = "";

  if (!name || !email || !phone || !zip || !address) {
    errorMsg.innerText = "Please fill out all fields.";
    return;
  }

  if (isNaN(phone)) {
    errorMsg.innerText = "Phone number must contain digits only.";
    return;
  }

  if (zip.length > 6) {
    errorMsg.innerText = "ZIP Code cannot be longer than 6 characters.";
    return;
  }

  if (!email.includes("@")) {
    errorMsg.innerText = "Please enter a valid email.";
    return;
  }

  calculateFinal(name, email);
}
```

Complete Your Donation

Full Name

Email Address

Phone Number (Numbers only)

ZIP Code (Max 6 chars)

No spaces or dashes.

Shipping Address

[Back to Shop](#)

[Confirm Donation](#)

Please fill out all fields.

D. Calculations (Discount & Tax)

The calculateFinal() function handles the math:

Subtotal: Loops through the cart array to sum up prices.

Discount: Checks if (cart.length >= 3). If true, it calculates a 10% deduction.

Tax: Calculates a % tax on the discounted subtotal.

Total: Subtotal - Discount + Tax.

All these values are dynamically inserted into the "Receipt" card on the final confirmation screen using document.getElementById().innerText.

Sources & References

Bootstrap 5 Documentation: Used for looking up component syntax (Navbar, Cards, Forms).

<https://getbootstrap.com/docs/5.0>

W3Schools JavaScript Reference: Learnt how to use some functions as isNaN, String Templates(template literals to insert HTML strings) & more.

Placeholder Images: Images used in the img/ folder are random stock images.